

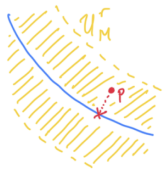
Reach of parametrised submanifolds

Daniel Platt (Imperial College London)
Thesis Family Journal Club, 27 May 2026

The reach is a quantity that measures how finely one needs to sample a submanifold of $\mathbb{R}^n$ to recover its geometry, roughly speaking. Much work has been done on computing (or bounding) it for zero sets of polynomials. In many applications one would like to know the reach for a submanifold that is given by a parametrisation, not as the zero set of a polynomial. In the talk, I will survey these applications and then explain one method to provably correctly bound the reach of such a parametrised subvariety.

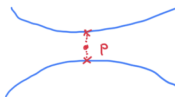
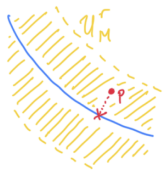
► $U_M^r := \{p \in \mathbb{R}^N : d(p, M) < r\}$

$\tau := \sup\{r \geq 0 : \text{for all } p \in U_M^r \text{ ex. unique nearest point on } M\}$



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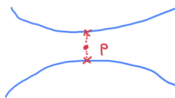
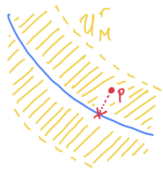
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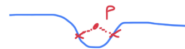
A **small bottleneck**
means small reach.

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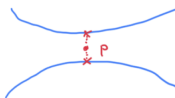
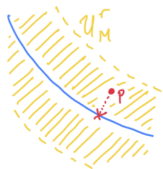
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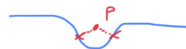
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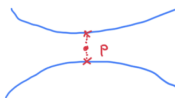
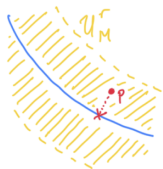
► Bottleneck: $\lambda := \min_{x \neq y, x-y \perp M} |x - y|$

► Radius of curvature at $p \in M$: $\rho(p) := \left(\max_{\substack{\gamma \text{ unit speed} \\ \text{geodesic with } \gamma(0)=p}} |\gamma''(0)|_{\mathbb{R}^N} \right)^{-1}$

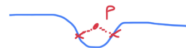
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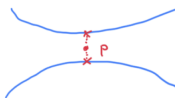
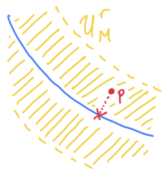
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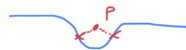
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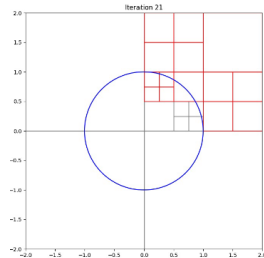
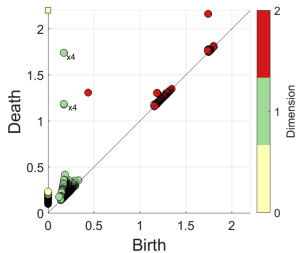
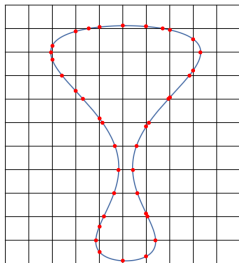
Approaches for varieties

- ▶ [AKC⁺19, ABL23]: given i.i.d. $\{p_i\} \subset M$, construct consistent **estimator** Θ_τ
- ▶ [DREG22, DREE⁺23]: find bottlenecks through **solving polynomial eqns**

$$\xi_1, \xi_2 \in \mathbb{R}, p, q \in \mathbb{R}^N : \quad f(p) = 0, \quad f(q) = 0 \\ p - q = \xi_1 \cdot \text{grad } f(p), \quad p - q = \xi_2 \cdot \text{grad } f(q).$$

Similar: find curvature maximum $\rightsquigarrow \tau$

- ▶ [DEHH19, DREG22]: get **dense sample** and **compute persistent homology**



- ▶ [PG25]: **branch-and-bound** for $\text{grad } f$ on $Z(f)$ $\rightsquigarrow \tau$

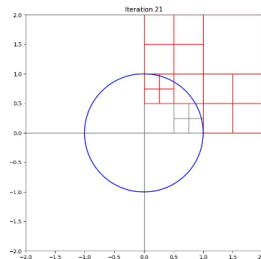
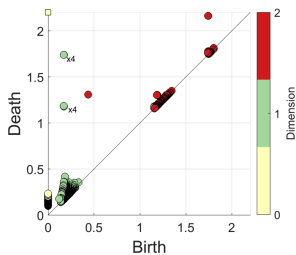
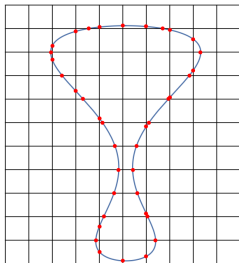
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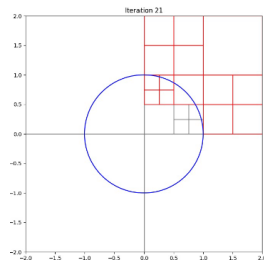
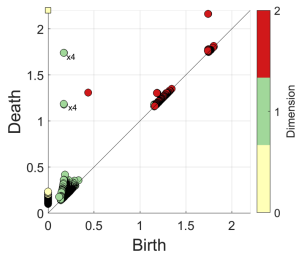
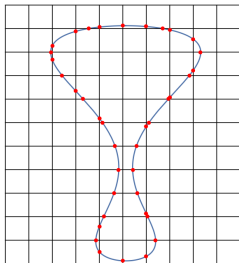
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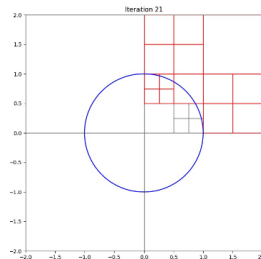
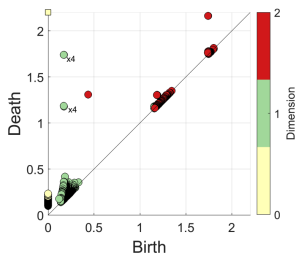
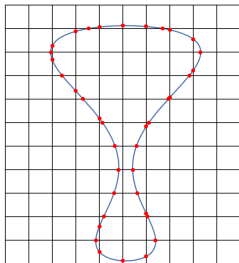
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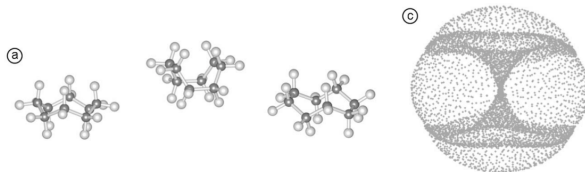
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Applications for varieties

- Applied maths: topology of kinematics **configuration space** [MTCW10]

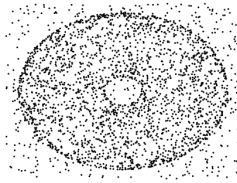


- **Pure maths:** proof ingredient [DPQB25]

$$\text{Quintic} := Z(g^* + \epsilon P) \subset \mathbb{CP}^4 \text{ for } \epsilon = \frac{27}{1000}, \quad (3.13)$$

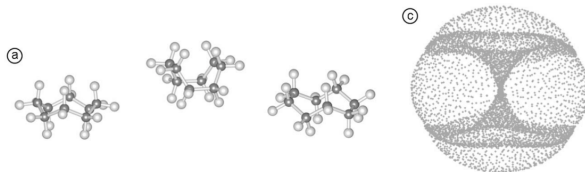
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- **Manifold reconstruction** (problem: use reach of the unknown manifold) [AL18]



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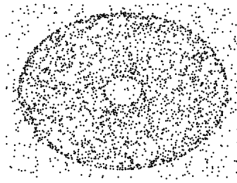


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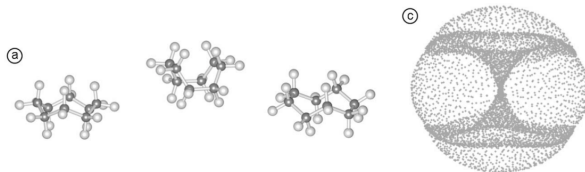
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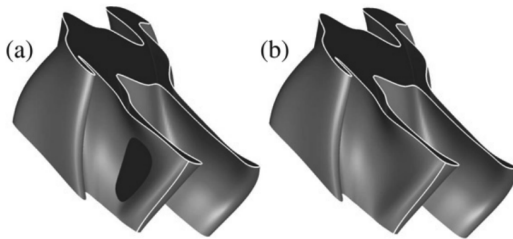
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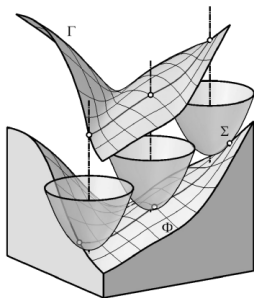


Applications for parametrised submanifolds

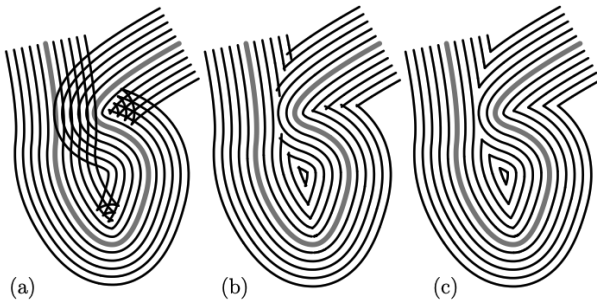
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- 2. **Millability** [PWGR98].

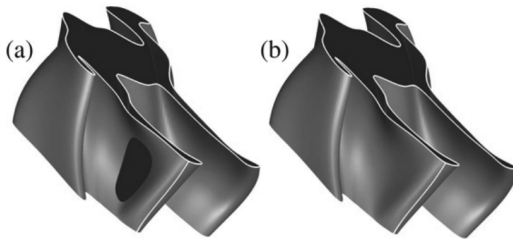


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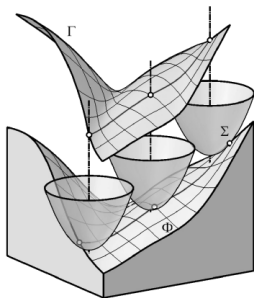


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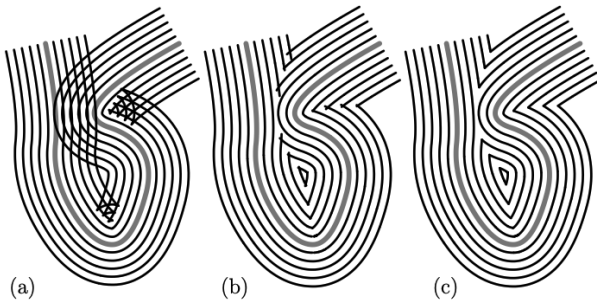
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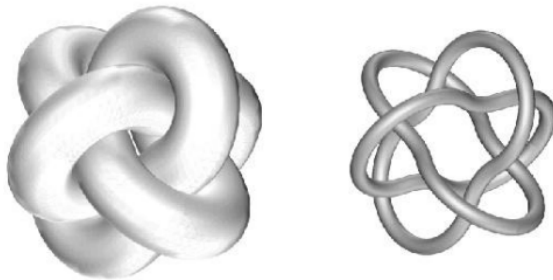


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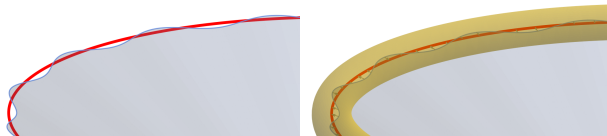


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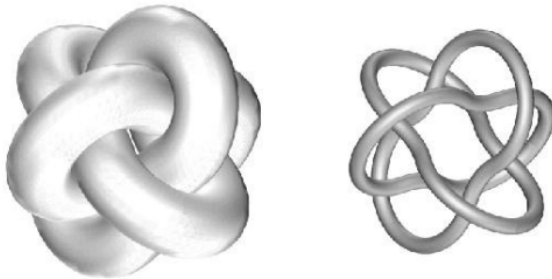


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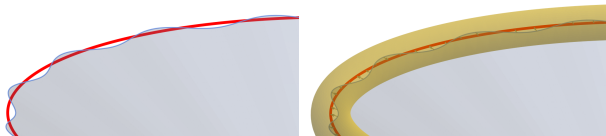


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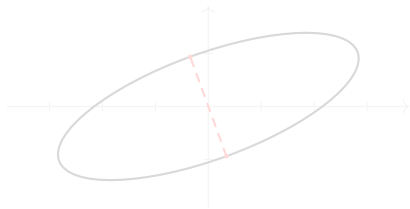


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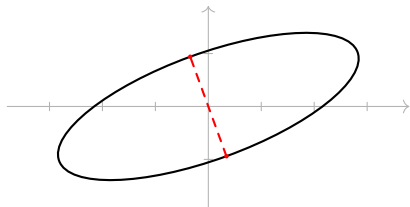
Approach for bounding the reach

- ▶ Let $\gamma : S^1 \rightarrow \mathbb{R}^2$
- ▶ Want to compute: $\tau = \min\{\lambda, \rho(p)\}$
- ▶ Step 1: compute **bottleneck size** λ
- ▶ $p = \gamma(t_1), q = \gamma(t_2)$ **bottleneck** if critical point for distance function
- ▶ E.g. $\gamma(t) = \begin{pmatrix} 3 \cos t \cos 20^\circ - \sin t \sin 20^\circ \\ 3 \cos t \sin 20^\circ + \sin t \cos 20^\circ \end{pmatrix}$
- ▶ $D(s, t) = \frac{1}{2} |\gamma(t) - \gamma(s)|_{\text{Eucl}}^2, \quad \mathcal{F}(s, t) = \nabla D(s, t)$
- ▶ $x_{\text{approx}} = (s_0, t_0) = (1.571, 4.712), \mathcal{E} = \mathcal{F}(x_{\text{approx}}) \rightsquigarrow |\mathcal{E}|_\infty \leq 2.1 \cdot 10^{-3}$
- ▶ $\mathcal{L} = d\mathcal{F}_{x_{\text{approx}}} = \text{Hess } D(x_{\text{approx}}), |\mathcal{L}^{-1}|_\infty \leq 0.501$
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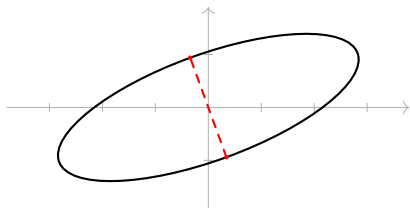
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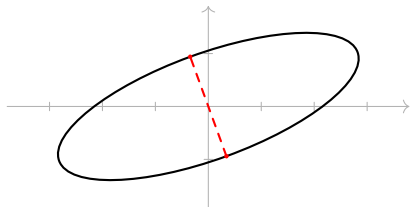
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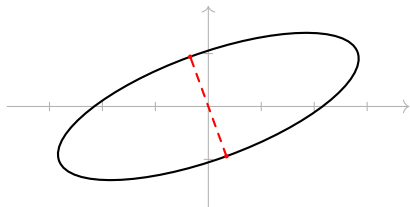
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- ▶ E.g. $\gamma(t) = \begin{pmatrix} 3 \cos t \cos 20^\circ - \sin t \sin 20^\circ \\ 3 \cos t \sin 20^\circ + \sin t \cos 20^\circ \end{pmatrix}$
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- ▶ $x_{\text{approx}} = (s_0, t_0) = (1.571, 4.712), \mathcal{E} = \mathcal{F}(x_{\text{approx}}) \rightsquigarrow |\mathcal{E}|_\infty \leq 2.1 \cdot 10^{-3}$
- ▶ $\mathcal{L} = d\mathcal{F}_{x_{\text{approx}}} = \text{Hess } D(x_{\text{approx}}), |\mathcal{L}^{-1}|_\infty \leq 0.501$
- ▶ Newton-Kantorovich $\Rightarrow$ **ex. exact bottleneck** x near x_{approx}



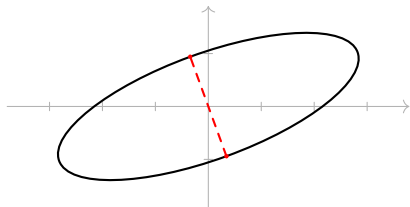
Approach for bounding the reach

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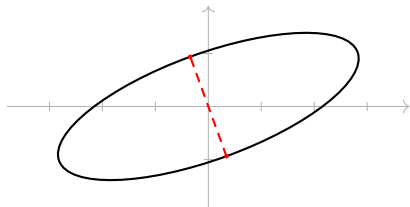
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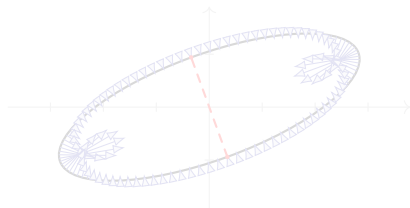


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- ▶ Let $\gamma(t) = \begin{pmatrix} 3 \cos t \cos 20^\circ - \sin t \sin 20^\circ \\ 3 \cos t \sin 20^\circ + \sin t \cos 20^\circ \end{pmatrix}$
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- ▶ Step 2: compute $\min_p \rho(p)$. For unit speed γ

$$\min_p \rho(p) = \min_{t \in S^1} \frac{1}{|\gamma''(t)|}$$

- ▶ Evaluate $|\gamma''(t)|$ at $t_0 = 0$, $t_1 = 0.01$, $t_2 = 0.02$, ... $\Rightarrow |\gamma''(t_i)| \leq 3$
- ▶ Compute **coarse bound** $|\gamma'''(t)| \leq C$, e.g. using $|\cos t| \leq 1$ get $|\gamma'''(t)| \leq 6$
 $\Rightarrow$ Bound everywhere! E.g.
 $|\gamma''(0.005)| \leq |\gamma''(0)| + |0.01 - 0.005| \cdot \max |\gamma'''| \leq 3 + 0.03$
- ▶ Conclusion: $\tau \geq \frac{1}{3.03}$

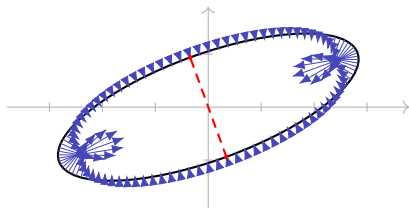


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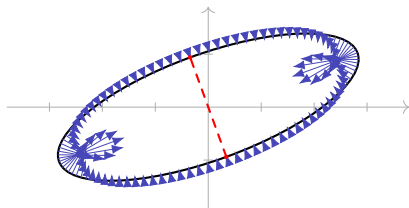


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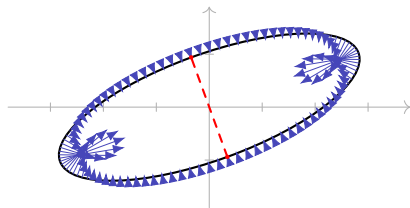


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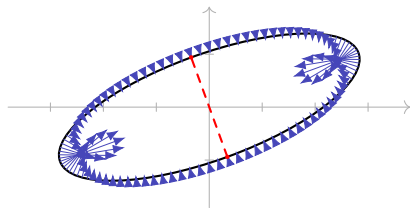


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Thank you for the attention!

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